

PHYSICS 7B – Spring 2025
Midterm 2, C. Bordel
Monday, March 31st, 7-9 pm

- Name:

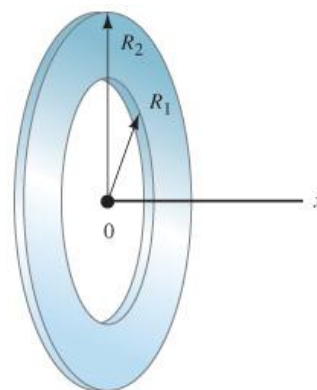
- Student ID #:

- Section #:

- **Rules:** *You may work on this exam from 7:10-9:10 pm PT. This midterm is closed book and closed notes and you are not allowed to use any electronic devices (computers, tablets, smartphones, calculators, ...). You are **not** allowed to communicate with any other students or discuss the content of the exam with anyone besides the Physics 7B teaching staff. Any violations of this policy will be considered a breach of academic integrity.*
- **Partial/Full Credit:** *Remember that you need to show your work and justify your answers in order to get full credit.*
- **Honors Statement:** *It is expected that during this examination, as with any examination that they undertake at the university, students adhere to the usual standards of academic integrity at the University of California at Berkeley as outlined on by the [Center for Teaching and Learning](#). Therefore, by submitting your exam, you are affirming the following statement:*
“I swear on my honor that I have neither given nor received aid on this exam. In addition, I abided by all of the examination policies as outlined above.”

Problem 1 - 25 points

A flat annulus of inner radius R_1 and outer radius R_2 carries a positive and uniform surface charge density σ . The origin of the symmetry axis (x -axis) is chosen at the center of the annulus, as shown in the figure.



- a- Determine the electric potential produced by the annulus at any point on the x -axis. Consider $V(\infty) = 0$. *Hint: it might be useful to consider a very thin ring first if you want to avoid a double integral.*

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- b-** Determine the direction of the electric field on axis and using part (a), determine its magnitude at any point on the symmetry axis.

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An electron of charge $-e$ and mass m is released at a small distance x from the center of the annulus described above. You may ignore the effect of gravity.

- c- Show that the electron undergoes simple harmonic motion about the center of the annulus. *Hint: one option is to use $\vec{F}_{net} = m \vec{a}$ to show that the electron's motion verifies $\frac{d^2 x}{dt^2} + \omega^2 x = 0$, which is characteristic of simple harmonic motion.*

- d- Based on the equation obtained in part (c), determine the angular frequency ω of the oscillatory motion.

Problem 2 – 25 points

Let's assume that an atom can be modeled as a positive point charge q carried by the nucleus located at the origin of a spherical coordinate system, surrounded by a volumetric distribution of negative charge with density $\rho_n(r) = -\frac{A}{r} \exp\left(\frac{-r}{a}\right)$ beyond the radial distance a representing an exclusion radius under which there are no negative charges. Note that r is the radial distance measured from the nucleus, $\exp$ the exponential function, and A is a positive constant to be determined.

- a-** Determine the constant A . *Hint: remember that an atom is electrically neutral.*

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b- Determine the electric field, in magnitude and direction, at a distance r from the nucleus. Consider $r > a$.

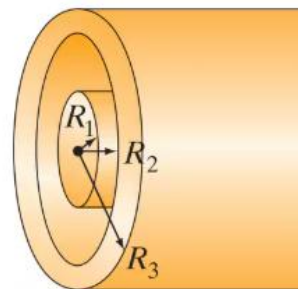
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c- Determine the electric potential at a distance r from the nucleus, with $r > a$. You may assume that $V(\infty) = 0$.

d- Sketch the electric field lines and equipotential lines with as much qualitative accuracy as possible regarding the spacing between the lines. Consider the range $r > a$.

Problem 3 – 25 points

A cylindrical core of radius R_1 is made of a non-conducting material carrying a uniform and positive volumetric charge density ρ_0 . Surrounding the solid cylinder is a coaxial cylindrical shell made of a neutral conducting material which extends from inner radius R_2 to outer radius R_3 . You may assume that the length L of the cylinder and shell is such that $L \gg R_3 > R_2 > R_1$.



- a-** Determine the electric charge per unit length of the cylindrical core.

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- b-** Determine the surface and volume charge densities - σ_{in} , σ_{out} and ρ_c - carried by the conducting cylindrical shell on its inner surface, outer surface and in volume, respectively. *Make sure you articulate your reasoning very clearly.*

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- c- Determine the direction and magnitude of the electric field at any radial distance from the symmetry axis. Consider $r < R_1$, $R_1 < r < R_2$, $R_2 < r < R_3$, and $r > R_3$.

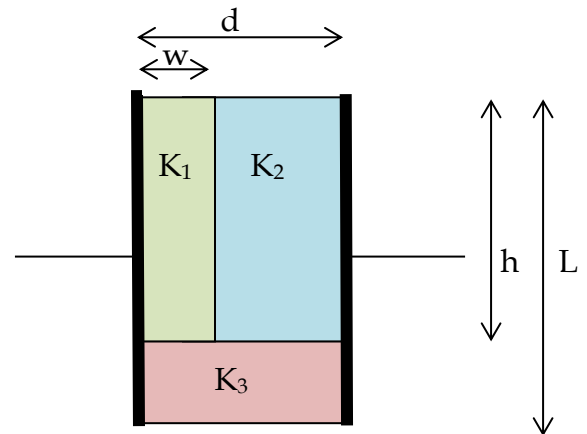
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- d-** Explain why the cylindrical core and cylindrical shell described above do not form a cylindrical capacitor.

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Problem 4 - 25 points

A parallel-plate capacitor made of square plates of side length L is filled with three different dielectric materials of dielectric constants K_1 , K_2 , and K_3 , as sketched in the figure (cross-section view). You may use all the lengths defined in the figure and assume that $d \ll L$. You may use the generic equation $C = K\epsilon_0 \frac{A}{d}$ without derivation.



- a- Draw the equivalent array of capacitors, each filled with only one type of material, with their dimensions.

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- b-** Using your diagram from part (a), determine the capacitance of this complex capacitor.

Electrostatics

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \hat{r} = k \frac{Q_1 Q_2}{r^2} \hat{r}$$

$$\vec{F} = q\vec{E}$$

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 r^2} \hat{r} = k \frac{dQ}{r^2} \hat{r}$$

$$\lambda = \frac{dQ}{dl} ; \quad \sigma = \frac{dQ}{dA} ; \quad \rho = \frac{dQ}{dV}$$

$$\vec{p} = Q\vec{d}$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$U = -\vec{p} \cdot \vec{E}$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

$$\Delta U = q\Delta V$$

$$V(b) - V(a) = - \int_a^b \vec{E} \cdot d\vec{l}$$

$$dV = \frac{dQ}{4\pi\epsilon_0 r} = k \frac{dQ}{r}$$

$$\vec{E} = -\vec{\nabla}V$$

$$\vec{F} = -\vec{\nabla}U$$

Capacitors

$$Q = CV$$

$$C_{eq} = C_1 + C_2 \text{ (In parallel)}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \text{ (In series)}$$

$$\epsilon = K\epsilon_0$$

$$U = \frac{Q^2}{2C}$$

$$u = \frac{\epsilon}{2} |\vec{E}|^2 \text{ (energy density)}$$

Gradient and displacement• **Cartesian coordinates :**

$$\vec{\nabla}f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$$

$$d\vec{r} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

• **Cylindrical coordinates :**

$$\vec{\nabla}f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{\partial f}{\partial z} \hat{z}$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + dz \hat{z}$$

• **Spherical coordinates :**

$$\vec{\nabla}f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

Lengths, Areas and Volumes

$$\text{Arc length: } dL = R d\theta$$

$$\text{Cylinder: } A = 2\pi RH ; V = \pi R^2 H$$

$$\text{Cylindrical shell: } dV = 2\pi R H dR$$

$$\text{Thin Circular Ribbon : } dA = 2\pi R dR$$

$$\text{Section of Thin Ribbon : } dA = R dR d\theta$$

$$\text{Sphere: } A = 4\pi R^2 ; V = \frac{4}{3} \pi R^3$$

$$\text{Spherical shell: } dV = 4\pi R^2 dR$$

Motion and Energy

$$\Delta K + \Delta U = W_{nc}$$

$$K = \frac{1}{2} m v^2$$

$$\vec{F}_{sp} = -k \vec{x} ; U_{sp} = \frac{1}{2} k x^2$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0 \text{ (simple harmonic motion)}$$

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Integrals

$$\int_0^{\infty} x^{2n} e^{-ax^2} dx = \frac{(2n)!}{n! 2^{2n+1}} \sqrt{\frac{\pi}{a^{2n+1}}}$$

$$\int_0^{\infty} x^{2n+1} e^{-ax^2} dx = \frac{n!}{2a^{n+1}}$$

$$\int x e^{-bx} dx = -\frac{e^{-bx}(bx+1)}{b^2}$$

$$\int (b + \frac{1}{x}) \frac{e^{-bx}}{x} dx = -\frac{e^{-bx}}{x}$$

$$\int (1+x^2)^{-1} dx = \arctan(x)$$

$$\int (1+x^2)^{-\frac{1}{2}} dx = \text{Ln}(\sqrt{1+x^2} + x)$$

$$\int x(b^2+x^2)^{-\frac{1}{2}} dx = \sqrt{b^2+x^2}$$

$$\int (1+x^2)^{-\frac{3}{2}} dx = \frac{x}{\sqrt{1+x^2}}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2)$$

$$\int \frac{1}{x} dx = \ln(x)$$

$$\int \frac{1}{\cos(x)} dx = \ln\left(\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right|\right)$$

$$\int \frac{1}{\sin(x)} dx = \ln\left(\left|\tan\left(\frac{x}{2}\right)\right|\right)$$

Taylor Expansions near $x=0$

$$\sin(x) \approx x$$

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

$$e^x \approx 1 + x + \frac{x^2}{2}$$

$$(1+x)^\alpha \approx 1 + \alpha x + \frac{(\alpha-1)\alpha}{2} x^2$$

$$\ln(1+x) \approx x - \frac{x^2}{2}$$

Trigonometry

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = 2 \cos^2(x) - 1$$

$$\sin(a+b) = \sin(a) \cos(b) + \cos(a) \sin(b)$$

$$\cos(a+b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$c^2 = a^2 + b^2 - 2ab \cos(\theta) \quad (\text{law of cosine})$$